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AI25BTECH11010 - Dhanush Kumar

Question:

The area enclosed between the straight line $y = x$ and the parabola $y = x^2$ in the xy -plane is (PI 2012)

1) $\frac{1}{6}$
2) $\frac{1}{4}$

3) $\frac{1}{3}$
4) $\frac{1}{2}$

Solution:

For $y = x^2$, we write it in matrix form as

$$\mathbf{x}^T V \mathbf{x} + 2\mathbf{u}^T \mathbf{x} + f = 0 \quad (1)$$

where

$$V = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad \mathbf{u} = \begin{pmatrix} 0 \\ -\frac{1}{2} \end{pmatrix}, \quad f = 0. \quad (2)$$

The line $y = x$ can be written parametrically as

$$\mathbf{x} = \mathbf{h} + \kappa \mathbf{m}, \quad \kappa \in \mathbb{R} \quad (3)$$

where

$$\mathbf{h} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \quad \mathbf{m} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}. \quad (4)$$

The intersection points satisfy

$$\kappa_i = \frac{-\mathbf{m}^T (V\mathbf{h} + \mathbf{u}) \pm \sqrt{[\mathbf{m}^T (V\mathbf{h} + \mathbf{u})]^2 - (\mathbf{m}^T V \mathbf{m}) g(\mathbf{h})}}{\mathbf{m}^T V \mathbf{m}} \quad (5)$$

where

$$g(\mathbf{h}) = \mathbf{h}^T V \mathbf{h} + 2\mathbf{u}^T \mathbf{h} + f. \quad (6)$$

Substituting values:

$$V\mathbf{h} + \mathbf{u} = \begin{pmatrix} 0 \\ -\frac{1}{2} \end{pmatrix}, \quad (7)$$

$$\mathbf{m}^T (V\mathbf{h} + \mathbf{u}) = -\frac{1}{2}, \quad (8)$$

$$\mathbf{m}^T V \mathbf{m} = 1, \quad (9)$$

$$g(\mathbf{h}) = 0. \quad (10)$$

Hence,

$$\kappa_{1,2} = \frac{-(-\frac{1}{2}) \pm \sqrt{(-\frac{1}{2})^2 - 1(0)}}{1} \quad (11)$$

$$= \frac{\frac{1}{2} \pm \frac{1}{2}}{1}. \quad (12)$$

Therefore,

$$\kappa_1 = 0, \quad \kappa_2 = 1. \quad (13)$$

The intersection points are

$$\mathbf{x}_1 = \mathbf{h} + \kappa_1 \mathbf{m} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \quad \mathbf{x}_2 = \mathbf{h} + \kappa_2 \mathbf{m} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}. \quad (14)$$

The area enclosed between the line and the parabola is

$$A = \int_0^1 (y_{\text{line}} - y_{\text{parabola}}) dx \quad (15)$$

$$= \int_0^1 (x - x^2) dx \quad (16)$$

$$= \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_0^1 \quad (17)$$

$$= \frac{1}{2} - \frac{1}{3} \quad (18)$$

$$= \frac{1}{6}. \quad (19)$$

$$\boxed{A = \frac{1}{6}} \quad (20)$$

Final Answer: (a) $\frac{1}{6}$